

School of Engineering

Project Tasks

(On – campus Classes)



ENG20009

Engineering Technology Inquiry Project

Semester 1 2026

Unit Convenor: A/Prof Rifai Chai

Industry:



Thermo Fisher Scientific Australia Pty Ltd

<https://www.thermofisher.com/au/en/home/>

Contact person:

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Tristan Duggan, Regional (VIC/TAS) & Channels Sales Manager

Thermo Fisher Scientific Inc. is an American supplier of scientific instrumentation, reagents and consumables, and software services. According to company figures, 46% of its sales are in life sciences, 20% in healthcare, and 34% in industrial/environmental and safety. One of their product is data loggers to be used as a monitoring device in various application for example in industry and environmental.

For interfacing with the end sensors, the two most popular communication protocol can be used are SDI12 and MODBUS. Learning to design protocol-based sensor and data logger are very much suitable in ENG20009, Engineering Technology Inquiry Project.

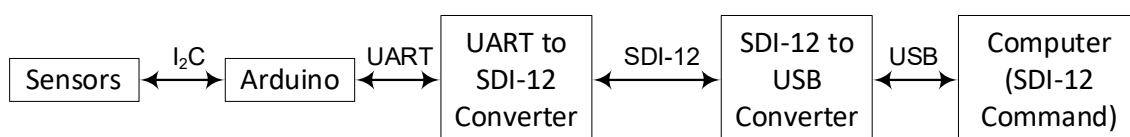
General Information:

This unit of study is a project-based unit in which students work in teams and aims to provide the skills to enquire and solve challenges oriented around engineering technologies.

For the portfolio project, you will work in a group of 5 students. Your team will be formed on the first week of your laboratory class. The project tasks will generally involve planning, design, construction, programming, testing and debugging. When you form the group, discuss allocation of these roles among your team members. Moreover, you should be actively involved in the discussion of each task design and fully understand the design details of your entire project, although some designs do not belong to your main role. It is advised that each team should choose a team coordinator to lead the team and chair group meetings.

1. Block Diagram of the Project

SDI12 Sensor



In the project task, students will work in a group consisting of 5 students to develop an SDI-12 sensor and data logger. The block diagram above shows the diagram of the SDI12 sensor

and interface that will be used in this unit. In addition, the diagram also shows the testing of the sensor from the computer. The components are started from the raw sensor with different types including:

- BME680 - Temperature, Humidity, Pressure and Gas Sensor or BME280- Temperature, Humidity, Pressure.
- BH1750FVI - Digital Light intensity Sensor

Each group use the two sensors and each sensor will be connected to the Arduino via the I2C. As soon as the data is received, this will be put together and transferred to the UART interface with the SDI commands capability.

For testing the sensor:

- For SDI12, we will need UART to SDI-12 and SDI-12 to USB converter. The USB can be connected to a computer and it's ready for testing with the SDI-12 commands.
- A serial port (COM) terminal emulation program can be used to test the SDI-12 commands from computer to the SDI-12 sensor. One of the terminal emulation program can be downloaded from :
<https://sites.google.com/site/terminalbpp/>

Pass: The "Compliant Sensor Node"

Objective: Implement the core SDI-12 logic to allow the Arduino to act as a "Slave" sensor.

Task: Create command parser that interprets serial strings based on the SDI-12 specification.

Requirements:

- **Polling Logic:** Continuously monitor the UART for commands.
- **Command Set:** Address Query, Change Address, Start Measurement, Send Data and Continuous Measurement.
- **Measurement:** When aM! is received, read the BME680 (Temp/Humidity) and BH1750 (Lux). Store the values in a buffer.
- **Data Delivery:** Respond to aD0! by returning the concatenated sensor values in standard format; aR0! is used for continuous measurements.

Name	Command	Response
Address Query	?!	a<CR><LF>
Change Address	aAb!	b<CR><LF>
Start Measurement	aM!	atttn<CR><LF>
Send Data	aD0! . . . aD9!	a<values><CR><LF> or a<values><CRC><CR><LF> a<values><CR><LF> or a<values><CRC><CR><LF> a<values><CR><LF> or a<values><CRC><CR><LF> a<values><CR><LF> or a<values><CRC><CR><LF> a<values><CR><LF> or a<values><CRC><CR><LF>
Continuous Measurements	aR0! ... aR9!	a<values><CR><LF>

A complete SDI-12 protocol/commands can be found at the following link:

https://www.sdi-12.org/current_specification/SDI-12_version-1_4-Jan-30-2021.pdf

Credit: The "Edge Data Hub"

Objective: Transform the node into a standalone logger that can function without a PC.

Task: Integrate a user interface and local persistence.

Requirements:

- **Local Logging:** Every 60 seconds, automatically save a timestamp (from RTC) and all sensor data to a .txt or .csv file on the **SD Card**.
- **Dashboard:** Display a "Real-time Dashboard" on the LCD showing current Temperature, Humidity, Pressure and Light Level.
- **Configuration Menu:** Use Pushbuttons to allow a user to manually trigger a log entry or clear the SD card memory without using the SDI-12 interface.

Distinction: "High-Performance Interrupt Architecture"

Objective: Replace polling with an interrupt-driven architecture to ensure no data loss.

Task: Shift critical I/O to Interrupt Service Routines (ISRs).

Requirements:

- **Timer-Based Sampling:** Use a Hardware Interrupt to sample the I2C sensors at precise intervals (e.g., exactly every 10ms for light intensity to calculate averages), ensuring the timing is independent of the main loop().
- **Visual Indicator:** Toggle an LED within the ISR to show when communication is active.

High Distinction: "Advanced Event-Driven Node"

Objective: Implement a professional-grade, power-efficient, and fail-safe system.

Task: Develop an Advanced Event-Driven Framework with enhanced data integrity.

Requirements:

- **State-Machine Parser:** Replace basic if/else command handling with a robust **Finite State Machine (FSM)**. This should handle partial or corrupted SDI-12 commands gracefully.
- Add a **"Watchdog Timer" (WDT)**. If the sensor logic hangs (e.g., an I2C sensor fails to respond), the WDT must automatically reset the Arduino to ensure the monitoring station never goes offline permanently.
- Applying new improvements to the project requirement (subject to the discussion with the workshop's facilitator)

Portfolio – Project Assessments:

Tasks and Details	Individual or Group	Weighting	Mapped Unit Learning Outcomes	Mapped Graduate Attributes	Assessment Due Date
Portfolio - Project i) Portfolio Project Brief (10%) ii) Portfolio Project Demonstration (35%) iii) Portfolio Project report (15%)	Group/ Individual Individual Group (Individual Contribution) Individual	60%	1,2,3,4,5,6,7,8	GA1, GA2, GA3, GA4, GA5, GA6	i) Apr 10, 2026 by 23:59pm (End of Week 5); submit in Canvas ii) Your on-campus workshop class in week 12 & project codes due in week 12. iii) Jun 05, 2025 by 23:59pm

(i) Project Brief

- Project planning and task division of each member. Particularly your individual tasks can be explained in more detail.
- Literature review on one of the applications of the SDI-12 sensor. The application of the SDI-12 sensor can be based on selected majors (electrical/electronics, software, robotic & mechatronics and software). This can be included a review of the hardware and software/firmware.
- The structure of the report includes the following:
 - Introduction (project planning and task division),
 - Main body of the review
 - Conclusion
 - References (IEEE style), please ensure there is in-text citation in the main body of the review.
- The report uses the IEEE, two-column format with a maximum of 4 pages. Template can be obtained using the following link:
<https://www.ieee.org/conferences/publishing/templates.html>.
- Brief to be submitted as a single PDF named "Surname, FirstName-Group #-PB".pdf

(ii) Project Demonstration

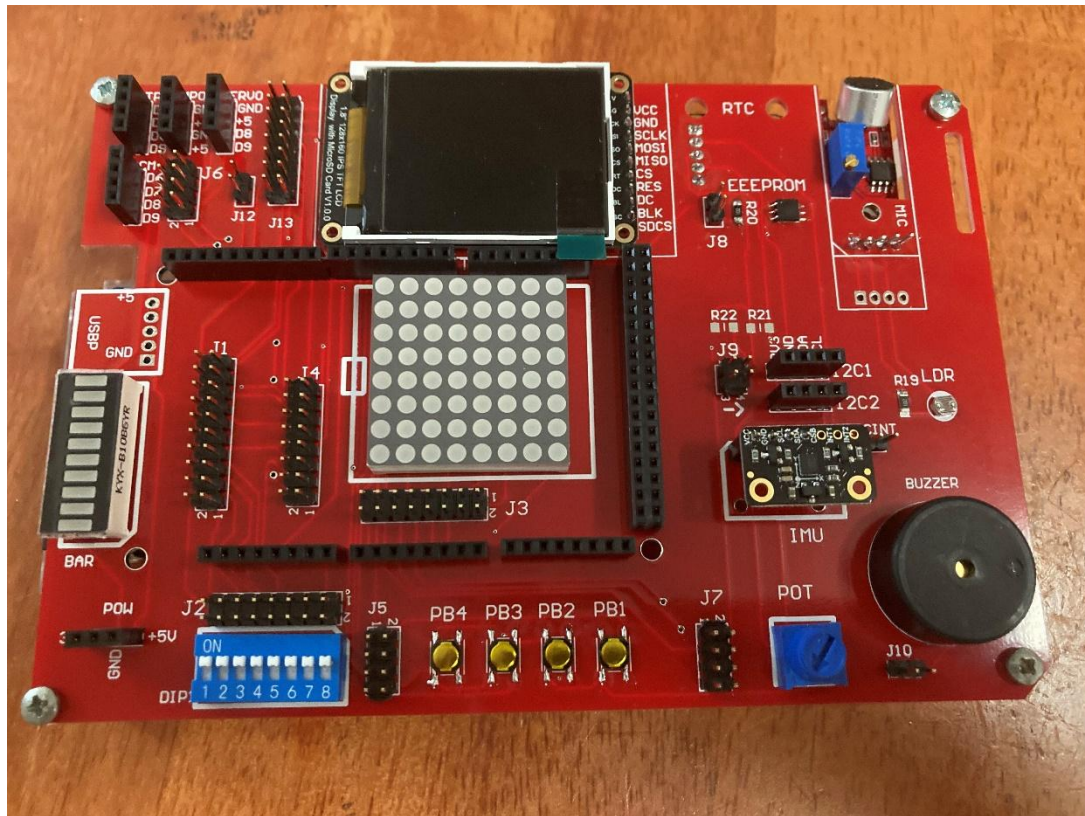
- To demonstrate project according to the design requirements.
- Each student needs to complete the Peer Review form by Week 12. Order of group members must be the same within each group's submission.
- One student in each group must submit a copy of project code demonstrated in class by Week 12.

- Code to be submitted as a single zip file named as “*Workshop Day-Time-Group #-Code*” .

(iii) Project Report

- Individual submission.
- Flowchart or pseudo code of the project tasks based on the task division.
This is only for the codes that you programmed and contributed.
- Risk Assessment for the application of the sensor.
Please use the information/provided with the General Risk Assessment (Week 8). You can choose to be applied to the project of this unit or one of the applications that you used in your project brief assignment.
- Reflection on knowledge learned in seminar, facilitation, lab and workshop for project work.
- Reflection on the teamwork according to the unit learning outcomes:
 - 1) How you function as an effective team member.
 - 2) How you communicated with the team, any stakeholders through appropriate verbal, written or technological communication to complete the project.
 - 3) Any outstanding work achieved, or major room for improvement, from yourself or the other team members.
- Report to be submitted as a PDF, named “*Surname, FirstName-Group #-PR*”.

Hardware - Project Board – Front View:



Hardware - Project Board – Back View (Arduino Due):



Hardware - Project Board – Schematic:

